

Homework 3 — SQL Fundamentals (Single-Table Queries)

SECTION A — ZAGI (5 Queries)

Question:

List the top 5 products by product price (columns: productid, productname, productprice) in descending price order.

SQL CODE:

-- Z1: Top 5 products by productprice

SELECT

productid,

productname,

productprice

FROM product

ORDER BY productprice DESC

LIMIT 5;

OUTPUT:

The screenshot shows a SQL IDE interface with a sidebar on the left displaying a database tree. The main window is titled 'Zagi/postgres@PostgreSQL 16' and contains a query editor with the following SQL code:

```
1 -- Z1: Top 5 products by productprice
2 SELECT
3     productid,
4     productname,
5     productprice
6 FROM product
7 ORDER BY productprice DESC
8 LIMIT 5;
9
```

Below the query editor, the 'Data Output' tab is active, showing the results of the query. The output is a table with three columns: productid, productname, and productprice. The first row is highlighted, showing productid 1, productname 'Garden of Eatin'', and productprice 1.99. The status bar at the bottom indicates 'Total rows: 0' and 'Query complete 00:00:01.755'.

productid	productname	productprice
1	Garden of Eatin'	1.99

Z2 — Categories With More Than 2 Products

Question:

Using only the product table, return categoryid values with more than 2 products.

SQL Code:

-- Z2: Count products per category; return those with > 2

SELECT

categoryid,

COUNT(*) AS product_count

FROM product

GROUP BY categoryid

HAVING COUNT(*) > 2

ORDER BY product_count DESC;

OUTPUT:

The screenshot shows a database management tool interface. On the left is a 'Server Explorer' pane showing a tree of servers and databases. The main area is split into two panes: 'Query' and 'Query History'. The 'Query' pane contains the following SQL code:

```
1  --- Z2: Count products per category; return those with > 2
2  SELECT
3      categoryid,
4      COUNT(*) AS product_count
5  FROM product
6  GROUP BY categoryid
7  HAVING COUNT(*) > 2
8  ORDER BY product_count DESC;
9
```

Below the query panes is a 'Data Output' pane. It shows the results of the query in a table with two columns: 'categoryid' (character (2)) and 'product_count' (bigint). The table is currently empty, with 'Total rows: 0' displayed at the bottom. The status bar at the bottom indicates 'Query complete 00:00:02.176' and 'LF Ln 9, Col 1'.

Z3 — Transactions Between 2013-01-02 and 2013-01-05

Question:

Count how many transactions occurred between '2013-01-02' and '2013-01-05' inclusive in sales transaction (column date). Return a single row with txn_count.

SQL Code:

```
-- Z3: Count transactions in date range
```

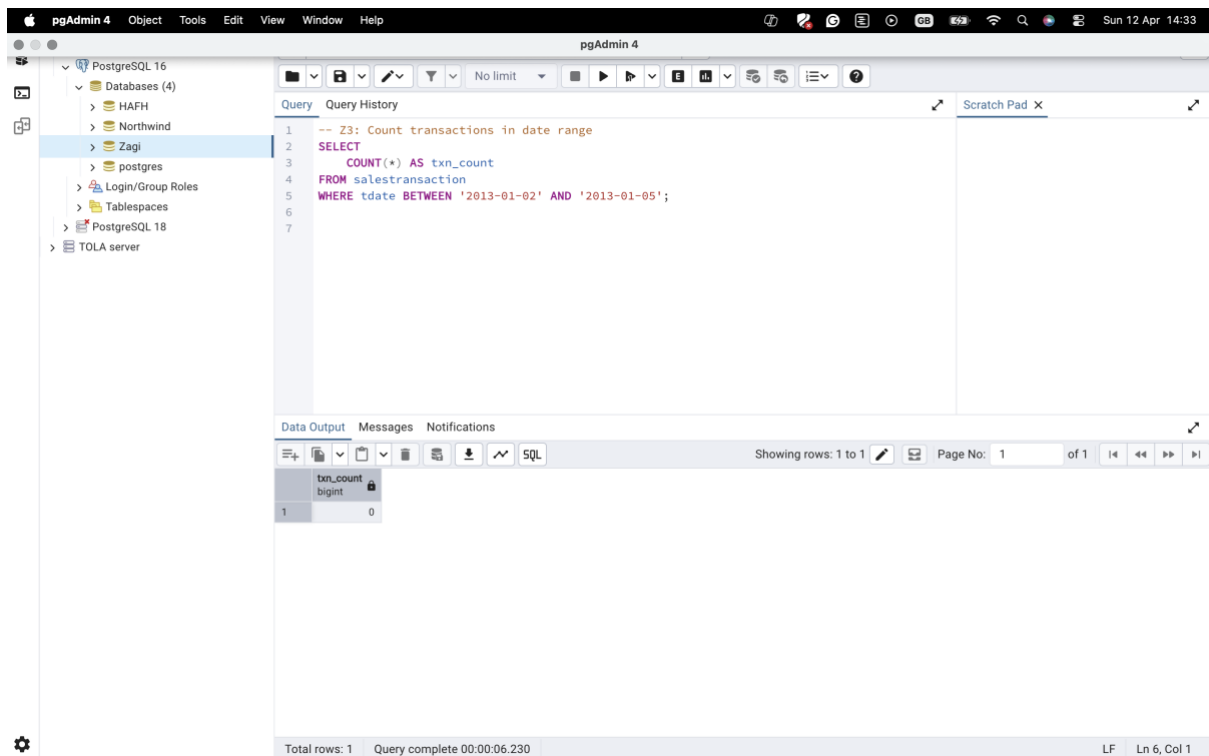
```
SELECT
```

```
    COUNT(*) AS txn_count
```

```
FROM salestransaction
```

```
WHERE tdate BETWEEN '2013-01-02' AND '2013-01-05';
```

OUTPUT:



The screenshot shows the pgAdmin 4 interface. The left sidebar displays a tree view of databases, with 'Zagi' selected. The main pane shows the SQL query editor with the following code:

```
1 -- Z3: Count transactions in date range
2 SELECT
3     COUNT(*) AS txn_count
4 FROM salestransaction
5 WHERE tdate BETWEEN '2013-01-02' AND '2013-01-05';
6
7
```

The 'Data Output' tab at the bottom shows the result of the query:

txn_count
0

The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:06.230'.

Z4 — Store Count Per Trimmed Region

Question.

For table store, return the number of stores per trimmed regionid as region, ordered by count desc—columns: region, store_count.

SQL Code:

-- Z4: Count stores per trimmed regionid

SELECT

 TRIM(regionid) AS region,

 COUNT(*) AS store_count

FROM store

GROUP BY TRIM(regionid)

ORDER BY store_count DESC;

OUTPUT:

The screenshot shows a database IDE interface. On the left is the Object Explorer showing a tree of servers and databases. The main pane displays the SQL query: `-- Z4: Count stores per trimmed regionid`, `SELECT`, `TRIM(regionid) AS region,`, `COUNT(*) AS store_count`, `FROM store`, `GROUP BY TRIM(regionid)`, `ORDER BY store_count DESC;`. Below the query editor is the Data Output pane, which shows the query results in a table with two columns: `region` (text) and `store_count` (bigint). The table is currently empty. A status bar at the bottom indicates 'Total rows: 0' and 'Query complete 00:00:03.516'.

region	store_count
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Z5 — Top 3 Products by Quantity Sold

Question:

From soldvia, return the top 3 productid by total quantity sold (noofitems). Columns: productid, total_qty. Order by total_qty desc.

SQL Code:

-- Z5: Sum quantity sold per product; return top 3

SELECT

productid,

SUM(noofitems) AS total_qty

FROM soldvia

GROUP BY productid

ORDER BY total_qty DESC

LIMIT 3;

OUTPUT:

The screenshot shows a PostgreSQL IDE interface. The top bar contains tabs for SQL, Statistics, Dependencies, Dependents, Processes, Dashboard, and two active connections: 'Zagi/postgres@Po...' and 'Zagi/postgres@PostgreSQL 16*'. Below the tabs is a toolbar with icons for file operations, query execution, and other functions. The main editor area displays the following SQL query:

```
1  -- Z5: Sum quantity sold per product; return top 3
2  SELECT
3      productid,
4      SUM(noofitems) AS total_qty
5  FROM soldvia
6  GROUP BY productid
7  ORDER BY total_qty DESC
8  LIMIT 3;
9
```

Below the query editor is a 'Data Output' tab, which is currently active. It shows the results of the query in a table format:

productid	total_qty
character (3)	bigint

At the bottom of the IDE, a status bar indicates 'Total rows: 0', 'Query complete 00:00:02.478', and the cursor position 'LF Ln 9, Col 1'.

SECTION B — HAFH (5 QUERIES)

H1 — Salary stats + count of managers with bonus

QUESTION:

Return AVG (msalary), MIN(msalary), MAX(msalary) as avg_salary, min_salary, max_salary, and COUNT (*) FILTER (WHERE mbonus IS NOT NULL) as with_bonus (or an equivalent).

SQL CODE:

-- H1: Salary stats + count of non-null bonuses

SELECT

AVG(msalary) AS avg_salary,

MIN(msalary) AS min_salary,

MAX(msalary) AS max_salary,

COUNT(*) FILTER (WHERE mbonus IS NOT NULL) AS with_bonus

FROM manager;

OUTPUT:

The screenshot shows a PostgreSQL query editor interface. The left sidebar displays a tree view of the database structure, including servers, databases, and tables. The main window shows a query being executed. The query is as follows:

```
-- H1: Salary stats + count of non-null bonuses
SELECT
  AVG(msalary) AS avg_salary,
  MIN(msalary) AS min_salary,
  MAX(msalary) AS max_salary,
  COUNT(*) FILTER (WHERE mbonus IS NOT NULL) AS with_bonus
FROM manager;
```

The query results are displayed in a table with the following columns: avg_salary, min_salary, max_salary, and with_bonus. The results show that the average salary is null, the minimum salary is null, the maximum salary is null, and the count of managers with bonuses is 0.

avg_salary	min_salary	max_salary	with_bonus
[null]	[null]	[null]	0

The bottom status bar indicates that the query was completed successfully and that there are 1 row(s) of data.

H2 — Buildings With ≥ 4 Floors

QUESTION:

List buildings with `bnooffloors` ≥ 4 , ordered by `bnooffloors` desc then `buildingid`. Columns: `buildingid`, `bnooffloors`.

SQL CODE:

- H2: Buildings with 4 or more floors

SELECT

`buildingid`,

`bnooffloors`

FROM `building`

WHERE `bnooffloors` ≥ 4

ORDER BY `bnooffloors` DESC, `buildingid`;

OUTPUT:

The screenshot shows a database management tool interface. On the left, the 'Object Explorer' pane displays a tree structure of databases, including 'PostgreSQL 16' and 'PostgreSQL 18'. The main window is titled 'HAFH/postgres@PostgreSQL 16' and contains a 'Query' editor with the following SQL code:

```
1  -- H2: Buildings with 4 or more floors
2  SELECT
3      buildingid,
4      bnooffloors
5  FROM building
6  WHERE bnooffloors >= 4
7  ORDER BY bnooffloors DESC, buildingid;
8
```

Below the query editor, the 'Data Output' pane shows the results of the query. It displays two columns: 'buildingid' (PK) character (3) and 'bnooffloors' integer. The results are currently empty, indicating that no data was returned by the query.

At the bottom of the interface, a status bar shows 'Total rows: 0' and 'Query complete 00:00:02.286'. The bottom right corner indicates 'LF Ln 8, Col 1'.

H3 — Industries With ≥ 3 Corporate Clients

QUESTIONS:

Count clients per cc industry. Return industries with $\text{COUNT}(*) \geq 3$, ordered by count desc.
Columns: ccindustry, client_count.

SQL CODE:

```
-- H3: Count clients per industry; return those with  $\geq 3$ 
```

```
SELECT
```

```
    ccindustry,
```

```
    COUNT(*) AS client_count
```

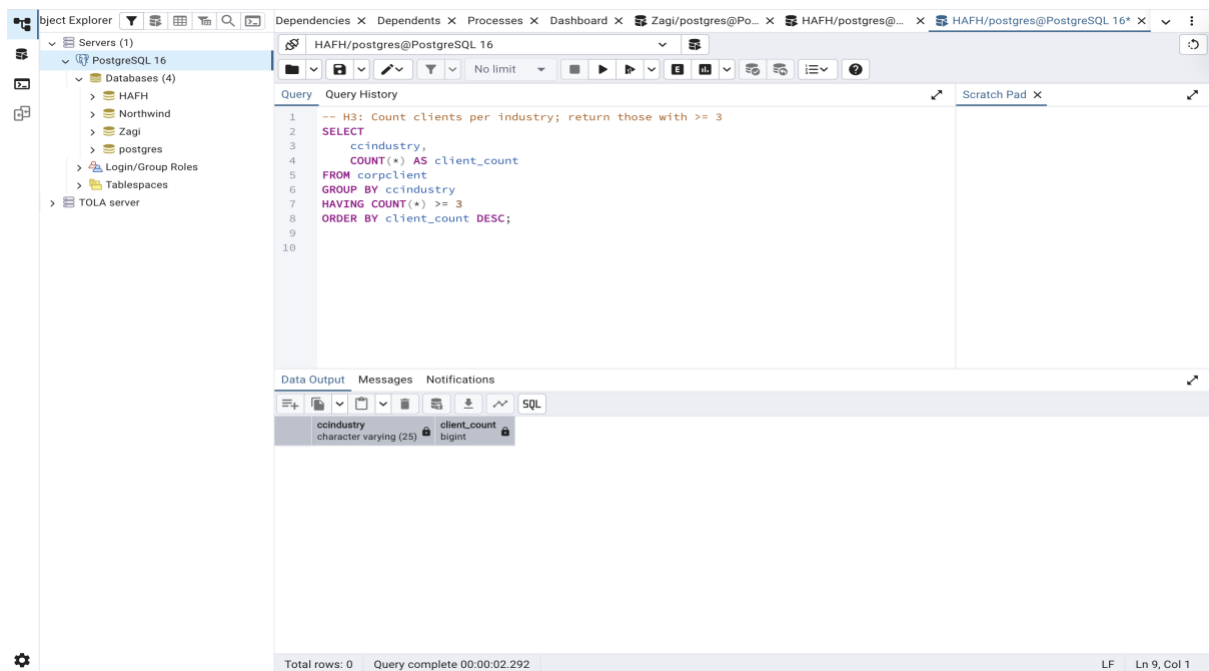
```
FROM corpclient
```

```
GROUP BY ccindustry
```

```
HAVING COUNT(*)  $\geq 3$ 
```

```
ORDER BY client_count DESC;
```

OUTPUT:



The screenshot shows a PostgreSQL IDE interface. On the left, the 'Object Explorer' pane displays the database structure, including 'Servers (1)', 'Databases (4)' (HAFH, Northwind, Zagi, postgres), 'Login/Group Roles', 'Tablespaces', and 'TOLA server'. The main window shows the 'Query' editor with the following SQL code:

```
-- H3: Count clients per industry; return those with  $\geq 3$ 
SELECT
    ccindustry,
    COUNT(*) AS client_count
FROM corpclient
GROUP BY ccindustry
HAVING COUNT(*)  $\geq 3$ 
ORDER BY client_count DESC;
```

Below the query editor, the 'Data Output' pane shows the results of the query. It displays two columns: 'ccindustry' (character varying (25)) and 'client_count' (bigint). The output is currently empty, indicating that the query has not yet been executed or the results are not visible.

At the bottom of the IDE, a status bar indicates 'Total rows: 0' and 'Query complete 00:00:02.292'.

H4 — Inspectors with ≥ 2 inspections scheduled in 2014

QUESTION:

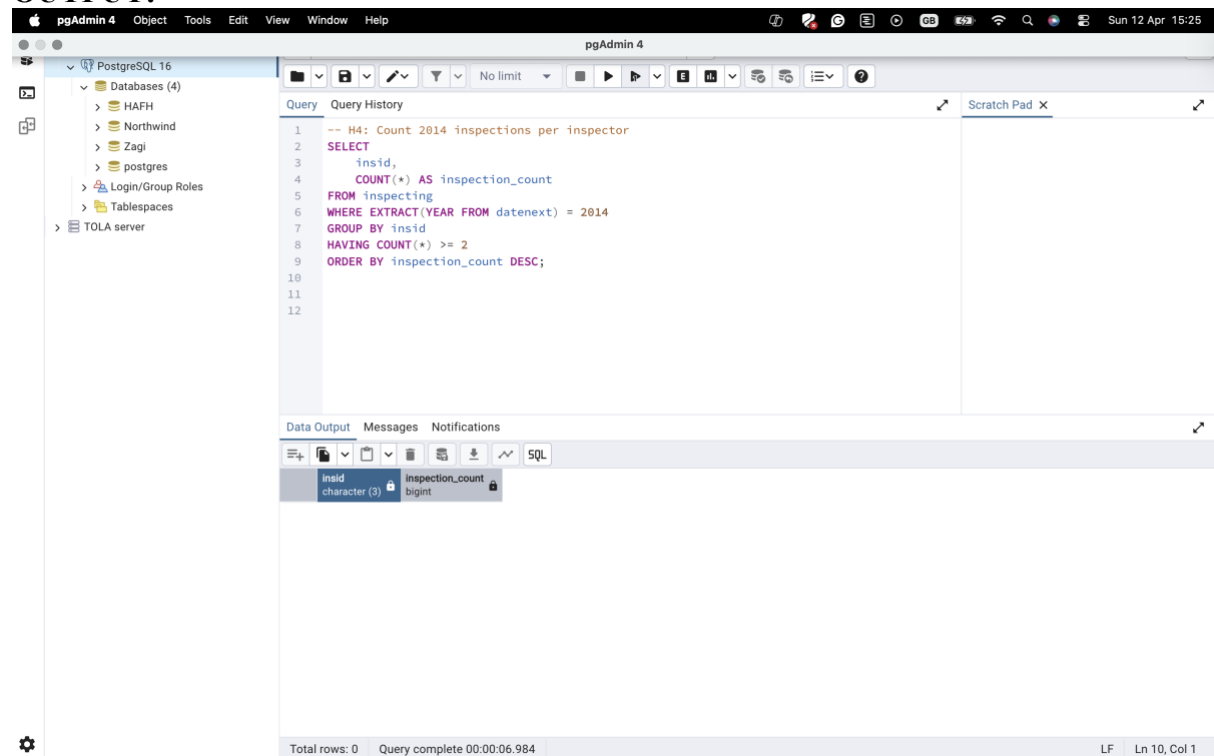
For rows where datenext is in 2014, return inspectors with at least 2 scheduled inspections.
Columns: insid, inspection_count, ordered by inspection_count desc.

SQL CODE:

```
-- H4: Count 2014 inspections per inspector

SELECT
    insid,
    COUNT(*) AS inspection_count
FROM inspecting
WHERE EXTRACT(YEAR FROM datenext) = 2014
GROUP BY insid
HAVING COUNT(*) >= 2
ORDER BY inspection_count DESC;
```

OUTPUT:



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure for PostgreSQL 16, including databases (HAFH, Northwind, Zagi, postgres), login/group roles, tablespaces, and the TOLA server. The main query editor window contains the following SQL code:

```
1  -- H4: Count 2014 inspections per inspector
2  SELECT
3      insid,
4      COUNT(*) AS inspection_count
5  FROM inspecting
6  WHERE EXTRACT(YEAR FROM datenext) = 2014
7  GROUP BY insid
8  HAVING COUNT(*) >= 2
9  ORDER BY inspection_count DESC;
```

The bottom panel shows the 'Data Output' tab with the following columns:

insid	inspection_count
character (3)	bigint

The status bar at the bottom indicates 'Total rows: 0' and 'Query complete 00:00:06.984'.

H5 — Managers with ≥ 2 phone numbers

QUESTION:

Count phone numbers per manager and return managers with $\text{COUNT}(*) \geq 2$. Columns: managerid, phone_count, ordered desc.

SQL CODE:

```
-- H5: Count phone numbers per manager; return those with  $\geq 2$ 
```

```
SELECT
```

```
    managerid,
```

```
    COUNT(*) AS phone_count
```

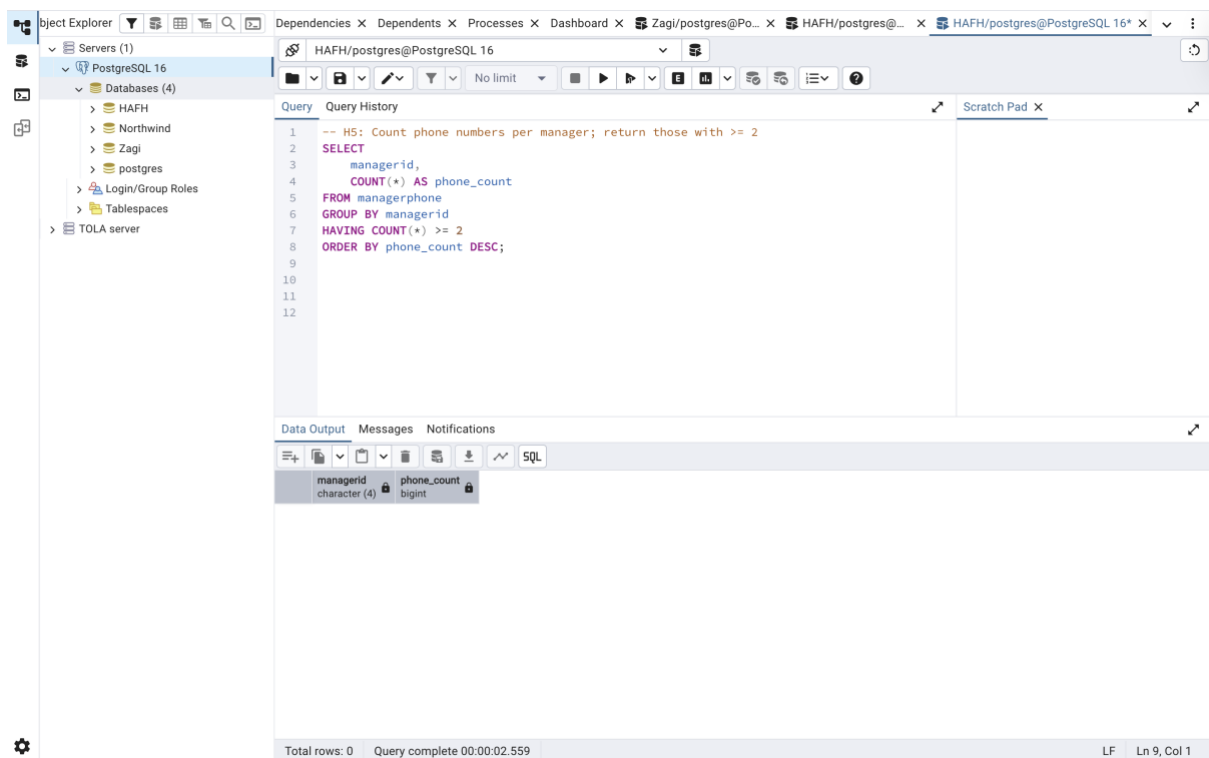
```
FROM managerphone
```

```
GROUP BY managerid
```

```
HAVING COUNT(*)  $\geq 2$ 
```

```
ORDER BY phone_count DESC;
```

OUTPUT:



The screenshot shows a PostgreSQL IDE interface. On the left, the 'Object Explorer' pane displays a tree view of the database structure, including 'Servers (1)', 'PostgreSQL 16', 'Databases (4)', and 'Tables (1)'. The main query editor displays the following SQL code:

```
-- H5: Count phone numbers per manager; return those with  $\geq 2$ 
SELECT
    managerid,
    COUNT(*) AS phone_count
FROM managerphone
GROUP BY managerid
HAVING COUNT(*)  $\geq 2$ 
ORDER BY phone_count DESC;
```

Below the query editor, the 'Data Output' pane shows the results of the query. The output is a table with two columns: 'managerid' (character (4)) and 'phone_count' (bigint). The table is currently empty, indicating that no results were returned for the given query.

The status bar at the bottom indicates 'Total rows: 0' and 'Query complete 00:00:02.559'.

SECTION C — NORTHWIND (5 QUERIES)

N1 — Top 10 most expensive active products

QUESTION:

“Which of our currently active products are the most expensive? Show the top 10 with their prices.” Analyst notes: Use products; filter active as discontinued = 0. Columns: productid, productname, unitprice. Sort by unitprice desc, then productname asc. Limit 10.

SQL CODE:

-- N1: Active products (discontinued = 0), top 10 by price

SELECT

productid,

productname,

unitprice

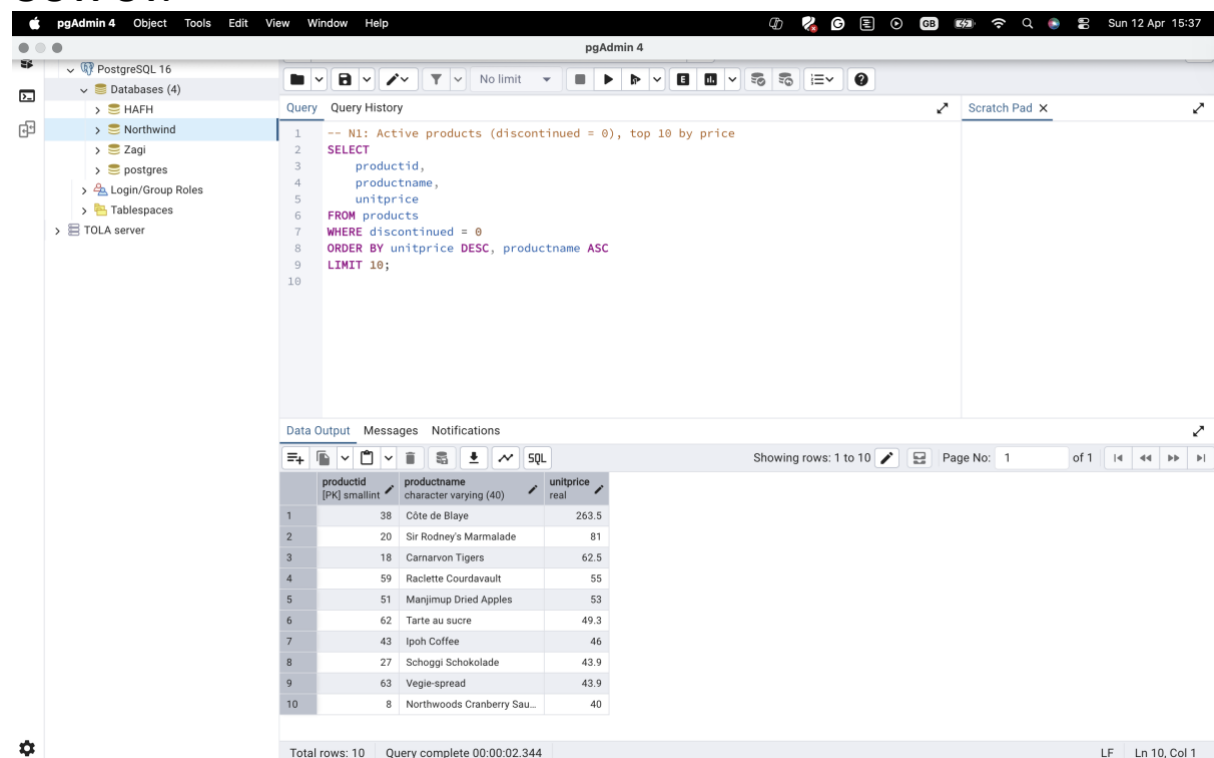
FROM products

WHERE discontinued = 0

ORDER BY unitprice DESC, productname ASC

LIMIT 10;

OUTPUT:



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including PostgreSQL 16, Databases (4), HAFH, Northwind, Zagi, postgres, Login/Group Roles, Tablespaces, and TOLA server. The main pane shows the SQL query for N1: Active products (discontinued = 0), top 10 by price. The query is as follows:

```
-- N1: Active products (discontinued = 0), top 10 by price
SELECT
  productid,
  productname,
  unitprice
FROM products
WHERE discontinued = 0
ORDER BY unitprice DESC, productname ASC
LIMIT 10;
```

The bottom pane shows the Data Output tab with the following results:

	productid [PK] smallint	productname character varying (40)	unitprice real
1	38	Côte de Blaye	263.5
2	20	Sir Rodney's Marmalade	81
3	18	Carnarvon Tigers	62.5
4	59	Raclette Courdavault	55
5	51	Manjimup Dried Apples	53
6	62	Tarte au sucre	49.3
7	43	Ispoh Coffee	46
8	27	Schoggi Schokolade	43.9
9	63	Vegie-spread	43.9
10	8	Northwoods Cranberry Sau...	40

The status bar at the bottom indicates: Total rows: 10, Query complete 00:00:02.344, LF, Ln 10, Col 1.

N2 — Top 5 customer countries

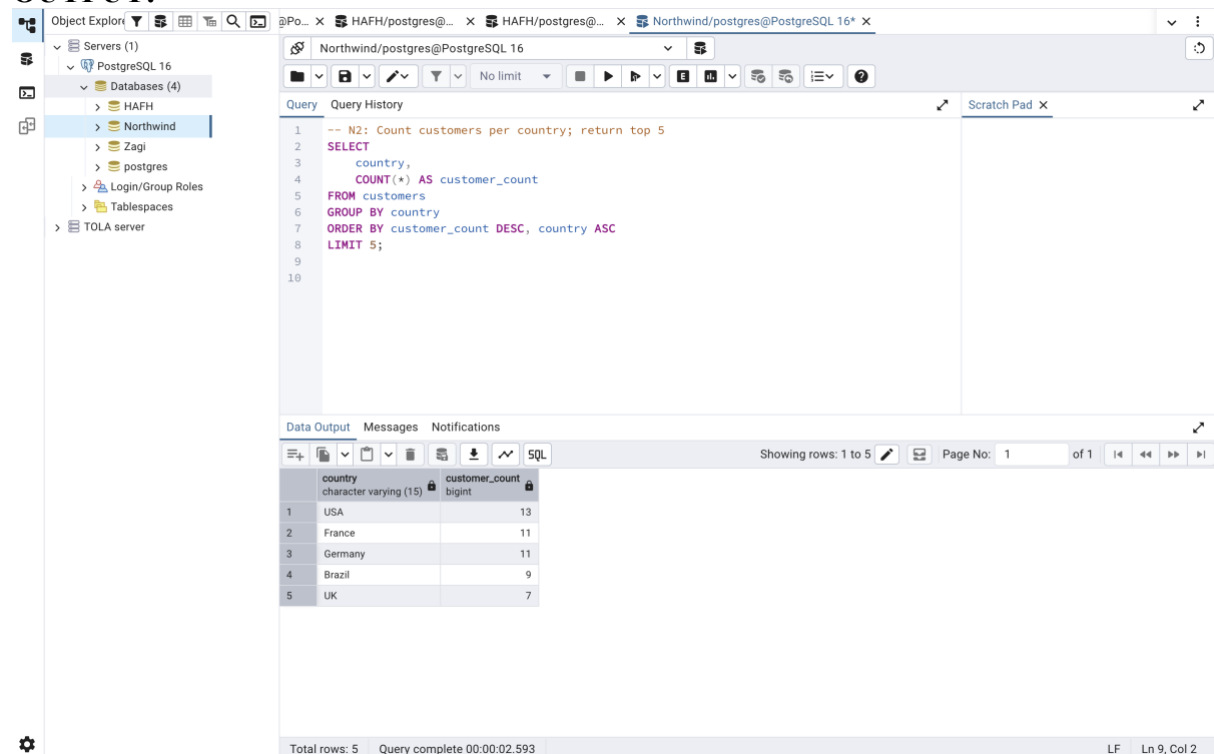
OUTPUT:

“Where are most of our customers located? Give me the top 5 countries by number of customers.” Analyst notes: Use customers. Columns: country, customer_count. Sort by customer_count desc, then country asc. Limit 5.

SQL CODE:

```
-- N2: Count customers per country; return top 5
SELECT
    country,
    COUNT(*) AS customer_count
FROM customers
GROUP BY country
ORDER BY customer_count DESC, country ASC
LIMIT 5;
```

OUTPUT:



The screenshot shows a PostgreSQL IDE interface. On the left is the Object Explorer showing a tree of databases including Northwind. The main window displays the SQL query from the previous block. Below the query editor, the Data Output tab is active, showing the results of the query. The results are displayed in a table with two columns: country and customer_count. The table shows the top 5 countries by customer count: USA (13), France (11), Germany (11), Brazil (9), and UK (7). The status bar at the bottom indicates 'Total rows: 5' and 'Query complete 00:00:02.593'.

	country	customer_count
1	USA	13
2	France	11
3	Germany	11
4	Brazil	9
5	UK	7

N3 — Number of orders received in 1996

Question:

“How many orders did we receive in the year 1996?” Analyst notes: Use orders. Return a single row column named orders_1996.

SQL CODE:

```
-- N3: Count orders from 1996
```

```
SELECT
```

```
    COUNT(*) AS orders_1996
```

```
FROM orders
```

```
WHERE EXTRACT(YEAR FROM orderdate) = 1996;
```

OUTPUT:

The screenshot shows a PostgreSQL IDE interface. On the left, the 'Object Explorer' pane displays a tree view of the database structure, including 'Servers (1)', 'PostgreSQL 16', 'Databases (4)', 'HAFH', 'Northwind', 'Zagi', 'postgres', 'Login/Group Roles', 'Tablespaces', and 'TOLA server'. The 'Northwind' database is selected. The main window displays the SQL query:

```
-- N3: Count orders from 1996
SELECT
    COUNT(*) AS orders_1996
FROM orders
WHERE EXTRACT(YEAR FROM orderdate) = 1996;
```

 The 'Data Output' pane at the bottom shows the result of the query: a single row with the column 'orders_1996' of type 'bigint' and the value '152'. The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:04.378'.

orders_1996
152

N4 — Top 5 products by line-item revenue

QUESTION:

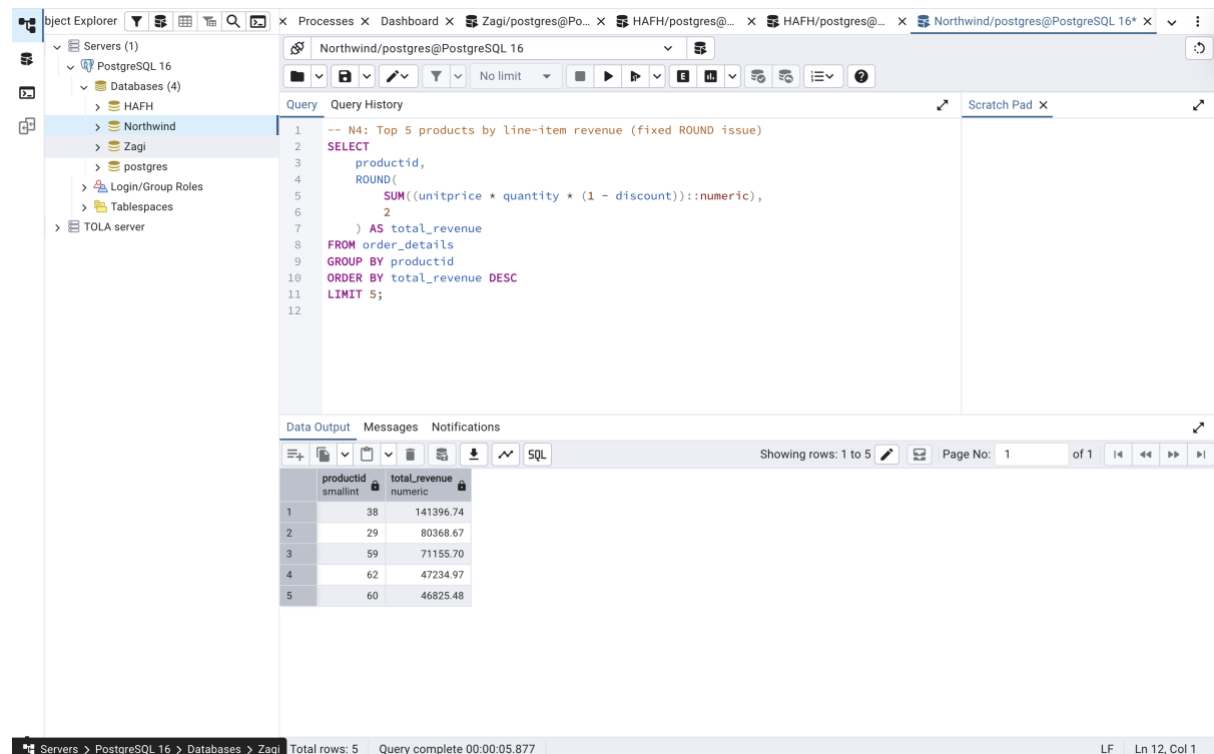
“Which products drive the most line-item revenue? Give me the top 5 product IDs by revenue.” Analyst notes: Use order_details. Revenue per product = unitprice * quantity * (1 - discount) aggregated by productid. Columns: productid, total_revenue (round to 2 decimals). Sort desc. Limit 5.

SQL CODE:

-- N4: Top 5 products by line-item revenue (fixed ROUND issue)

```
SELECT
    productid,
    ROUND(
        SUM((unitprice * quantity * (1 - discount))::numeric),
        2
    ) AS total_revenue
FROM order_details
GROUP BY productid
ORDER BY total_revenue DESC
LIMIT 5;
```

OUTPUT:



The screenshot shows a PostgreSQL IDE interface. The left sidebar displays the 'Object Explorer' with a tree view of databases: PostgreSQL 16, HAFH, Northwind, Zagi, postgres, Login/Group Roles, Tablespaces, and TOLA server. The main window shows the 'Query' editor with the following SQL code:

```
-- N4: Top 5 products by line-item revenue (fixed ROUND issue)
SELECT
    productid,
    ROUND(
        SUM((unitprice * quantity * (1 - discount))::numeric),
        2
    ) AS total_revenue
FROM order_details
GROUP BY productid
ORDER BY total_revenue DESC
LIMIT 5;
```

Below the query editor, the 'Data Output' tab is active, displaying the results of the query in a table format. The table has two columns: 'productid' (smallint) and 'total_revenue' (numeric). The results are as follows:

productid	total_revenue
38	141396.74
29	80368.67
59	71155.70
62	47234.97
60	46825.48

The status bar at the bottom indicates 'Total rows: 5' and 'Query complete 00:00:05.877'.

N5 — Employees hired in 1993

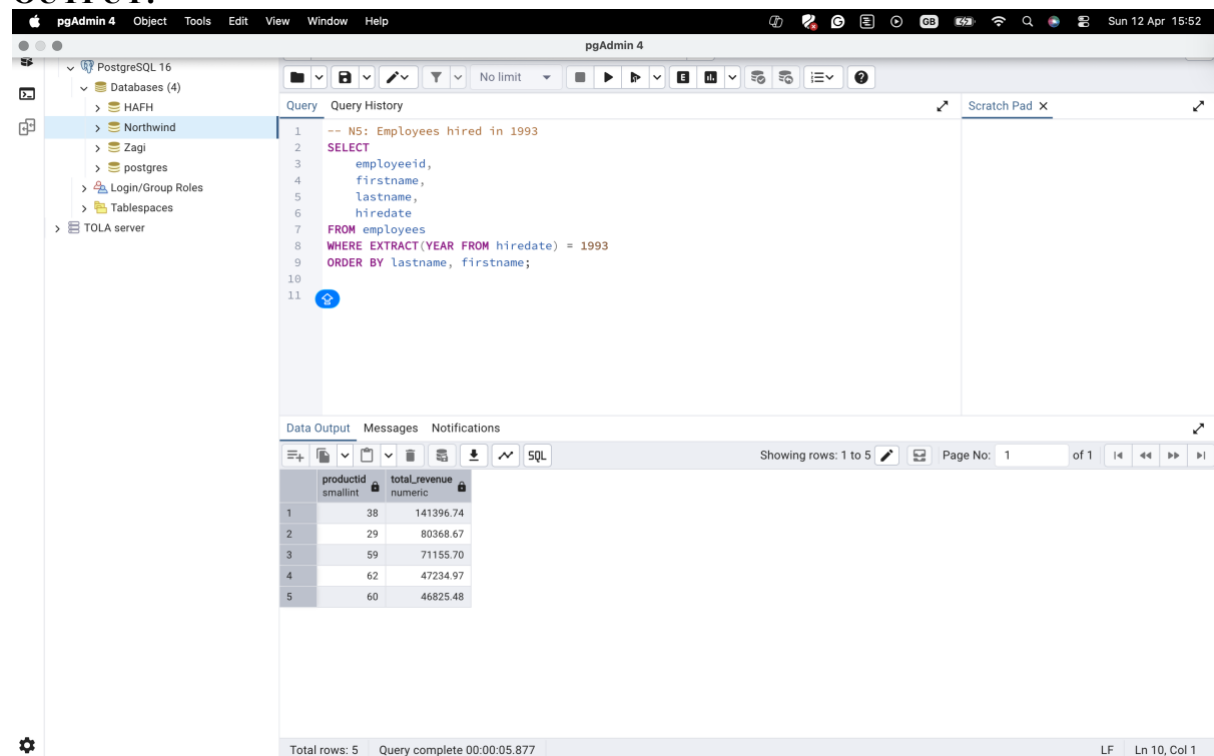
QUESTION:

“Who joined the company in 1993? List their names and hire dates.” Analyst notes: Use employees—columns: employeeid, firstname, lastname, hiredate. Sort by lastname, then firstname.

SQL CODE:

```
-- N5: Employees hired in 1993
SELECT
    employeeid,
    firstname,
    lastname,
    hiredate
FROM employees
WHERE EXTRACT(YEAR FROM hiredate) = 1993
ORDER BY lastname, firstname;
```

OUTPUT:



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including PostgreSQL 16, Databases (4), HAFH, Northwind, Zagi, postgres, Login/Group Roles, Tablespaces, and TOLA server. The main pane shows the SQL query:

```
-- N5: Employees hired in 1993
SELECT
    employeeid,
    firstname,
    lastname,
    hiredate
FROM employees
WHERE EXTRACT(YEAR FROM hiredate) = 1993
ORDER BY lastname, firstname;
```

The bottom pane shows the Data Output tab with the following results:

	productid smallint	total_revenue numeric
1	38	141396.74
2	29	80368.67
3	59	71155.70
4	62	47234.97
5	60	46825.48

The status bar at the bottom indicates "Total rows: 5" and "Query complete 00:00:05.877".